Problem

Title

Reverse Only Even Digits of a Number

Topics

Digit Extraction, Conditional Statements, Loops, Number Manipulation, Mathematical Reconstruction

Difficulty Level

Hard

Problem Detailed Statement (Scenario Based)

In a logic-building coding session, students are given a challenging number manipulation task.

The trainer provides a positive integer **N**. Students must reverse **only the even digits** of the number while keeping the odd digits in their original positions.

Rules:

1. Extract all digits of the number.
2. Identify the even digits.
3. Reverse the sequence of only the even digits.
4. Place them back into their original positions.
5. Odd digits must remain unchanged.

For example:

If N = 123456

Digits: 1 2 3 4 5 6
Even digits: 2 4 6
Reversed even digits: 6 4 2

Final number:
1 6 3 4 5 2

Output: **163452**

Another example:

If N = 2468
Even digits: 2 4 6 8
Reversed: 8 6 4 2
Output: **8642**

If N = 13579
There are no even digits → Output remains same.

Students must:

- Use loops for digit extraction.
- Avoid converting number into string.
- Carefully reconstruct the final number mathematically.
- Handle edge cases such as:
    - Single digit numbers
    - Numbers with no even digits
    - Large input values

This problem strengthens:

- Selective digit manipulation
- Conditional filtering
- Reverse logic
- Mathematical reconstruction
- Loop-based implementation

Input Format

A single integer:

```
N
```

Constraints:

```
0 ≤ N ≤ 10^12
```

Output Format

- Print the modified number after reversing only the even digits.

Examples

Example 1

Input:

```
123456
```

Output:

```
163452
```

Example 2

Input:

```
2468
```

Output:

```
8642
```

Example 3

Input:

```
13579
```

Output:

```
13579
```

Example 4

Input:

```
1020304
```

Output:

```
1040302
```